ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 D ✓ 1.1.4 A ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 I ✓ 1.3.3 G ✓ 1.3.4 B ✓ 1.3.5 E ✓ 1.3.6 A ✓ 1.3.7 F ✓ 1.3.8 H ✓ 1.3.9 J ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A+ B+ B− A− ✓: A extends (clamp), B extends (drill down), B retracts (drill up), A retracts (unclamp) ✓ (2)

2.2 Two lines (A and B) against steps 1 to 5 ✓ with positions 0 and 1 labelled ✓. A: rises from 0 to 1 between steps 1 and 2 ✓, stays at 1 until step 4, falls to 0 between steps 4 and 5 ✓. B: stays at 0 until step 2, rises to 1 between steps 2 and 3 ✓, falls to 0 between steps 3 and 4 ✓. (6)

2.3 a0 – clamp A is fully retracted (open) ✓; a1 – A is fully extended (part clamped) ✓; b0 – drill cylinder B is fully up (retracted) ✓; b1 – B is fully down (hole drilled to depth) ✓ (4)

2.4 Any THREE ✓✓✓: two-hand start or light curtain; fixed / interlocked guard around the drill; emergency-stop button; drill cannot start unless the part is clamped (a1); safe exhaust / pressure release when stopped; warning signs and lights. (3)

2.5 $A = F \div P = 800 \div 600\,000 = 1{,}33 \times 10^{-3}\ m^2$ ✓; $D = \sqrt{(4A \div \pi)} = \sqrt{(4 \times 1{,}33 \times 10^{-3} \div \pi)} = 0{,}0412\ m = 41{,}2\ mm$ ✓; choose 50 mm (the next size up) ✓ (3)

[18]

QUESTION 3

3.1 When the solenoid is energised the valve shifts and sends air to the rear of the cylinder, so it extends ✓; when the solenoid is de-energised the spring returns the valve and air goes to the front, so the cylinder retracts ✓. (2)

3.2 Inputs ✓✓: Start – I0.0; a0 – I0.1; a1 – I0.2; b0 – I0.3; b1 – I0.4. Outputs ✓: solenoid A+ (clamp) – Q0.0; solenoid B+ (drill down) – Q0.1. Neat table with description and address columns ✓. (Any consistent addresses.) (4)

3.3 Start (NO) ✓ with a seal-in Motor contact in parallel ✓; in series with Stop (NC) ✓, guard switch contact that opens when the guard is lifted ✓ and overload (NC), driving the Motor coil ✓, e.g. (5)

```
|--+--[ Start ]--+--[/ Stop ]--[ Guard closed ]--[/ OL ]--( Motor )--|
|  |             |
|  +--[ Motor ]--+
```

3.4 When b1 (drill at the bottom) is reached it starts the TON timer, set to 3 s ✓; the B+ output stays on while the timer counts ✓; when the timer's done bit switches on, the program switches off B+ so that B retracts ✓. (3)

3.5 NPN (sinking): the sensor switches 0 V (negative) to the load / PLC input ✓; used with a sourcing PLC input card ✓. PNP (sourcing): the sensor switches +24 V to the input ✓; used with a sinking PLC input card ✓. The sensor type must match the input card. (4)

[18]

QUESTION 4

4.1 Sensor ✓ and reason ✓ for each: (a) inductive proximity sensor – detects metal without contact, unaffected by dirt and oil; (b) capacitive proximity sensor – detects non-metals such as plastic; (c) through-beam photoelectric sensor – long range across the conveyor, detects any opaque object; (d) reed switch / magnetic cylinder sensor – detects the magnet on the piston through the cylinder wall. (8)

4.2 $S_{stainless} = 8 \times 0{,}7 \checkmark = 5{,}6 \text{ mm} \checkmark$ (2)

4.3 Analogue: output varies continuously with the quantity (e.g. 4–20 mA) ✓, e.g. pressure transducer on the air supply / temperature sensor on the drill motor ✓. Digital: output is only ON or OFF ✓, e.g. limit switch, inductive sensor at the clamp ✓. (4)

4.4 Hysteresis is the difference between the distance at which the sensor switches on and the distance at which it switches off ✓; it stops the output from switching on and off rapidly (chattering) when the object is right at the edge of the range ✓. (2)

[16]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: emergency-stop button; fixed or interlocked guard around the drill; two-hand start / light curtain; drill cannot run unless the part is clamped; soft-start / pressure-dump valve; warning lights and signs; chip guard / shield. (4)

5.2 In order ✓✓✓✓: tell people and stop the machine normally; isolate the energy sources (switch off and lock the electrical isolator, close the air valve and bleed off the pressure); fit your own lock and a danger tag; try to start the machine to prove it is dead; then do the work (and remove the lock only when finished). (4)

5.3 Any FOUR ✓✓✓✓: air pressure too low / regulator set wrong; flow-control valve closed too far; leaking or worn piston seal; leaking pipes or fittings; blocked filter or silencer; sticking valve; kinked or too-small air pipe; mechanical binding / lack of lubrication. (4)

5.4 Any FOUR checks ✓✓✓✓: drain water from the filter bowl; check the regulator pressure on the gauge; check the oil level in the lubricator; listen for and fix air leaks; check pipes and fittings for damage; check that cylinders and limit switches are tight; test the emergency stop. (4)

5.5 Any ONE duty ✓✓: provide and maintain a safe workplace and safe machines; provide training, information and supervision; provide protective equipment; report incidents; carry out risk assessments. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE